6.1 ESP32-Cam Introduction

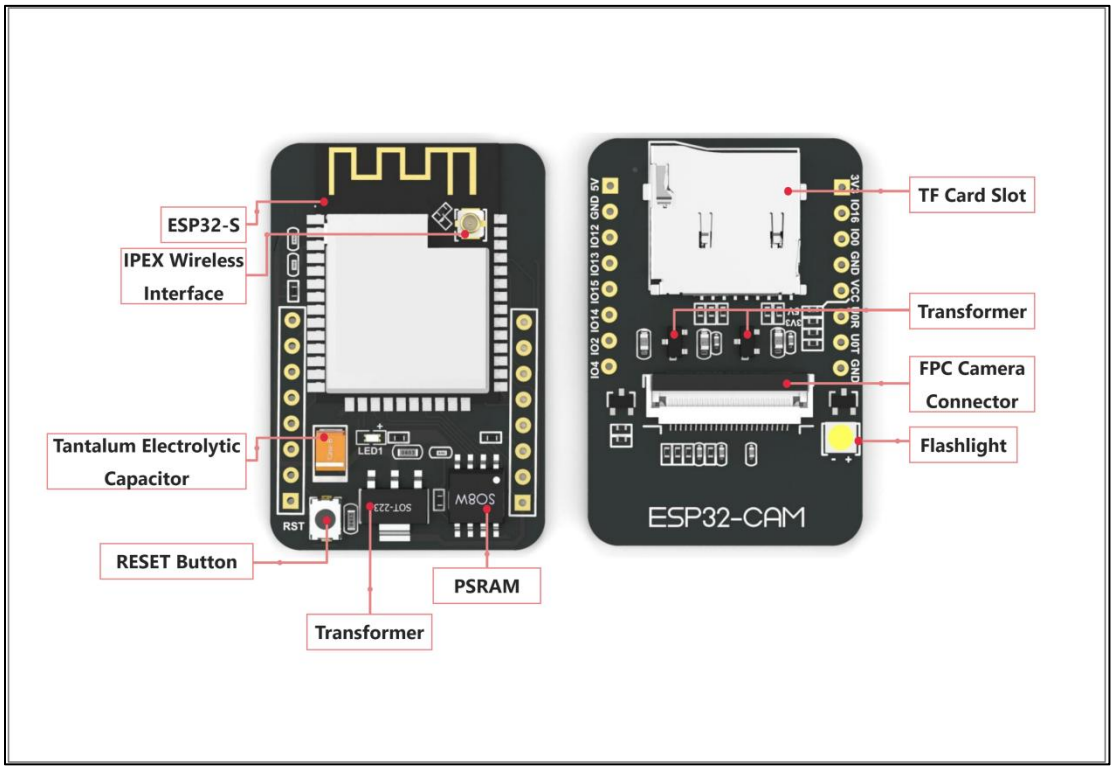
1. ESP32-Cam Instruction

1.1 Hardware

ESP32-CAM is a compact camera module based on ESP32 microcontroller. It can independently run as the minimum system with dimensions of only 27*40.5*4.5mm and a minimum deep-sleep current of 6mA.

The ESP32-CAM adopts a DIP package, which can be directly plugged onto the baseboard for convenient and reliable connection. This design makes it suitable for various IoT hardware terminal applications.

The hardware construction is shown below:



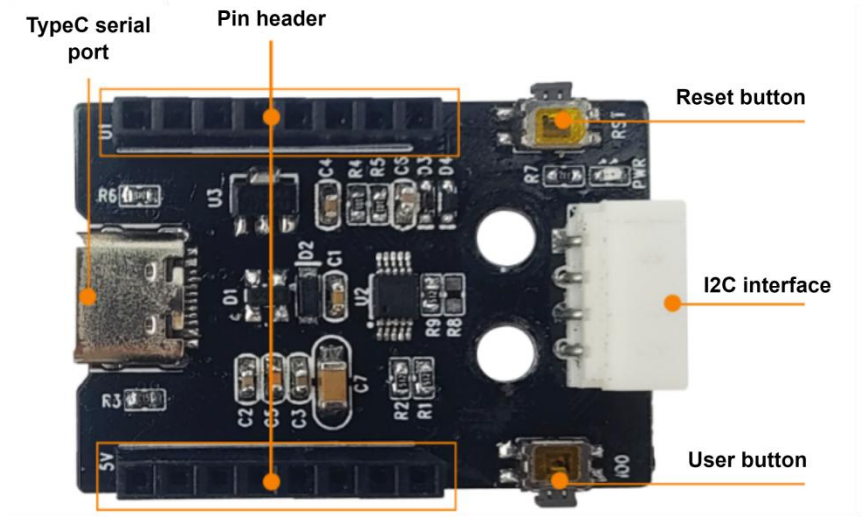
1.2 Working Principle

ESP32-cam module utilizes the built-in camera module to capture images.

The ESP32 microcontroller then processes the image data and facilitates wireless transmission via the Wi-Fi module or storage onto a microSD card. With support for various communication protocols and low-power features, it finds extensive application across diverse IoT scenarios.

2. Interface Instruction and Installation

2.1 Camera Expansion Board Interface Instruction

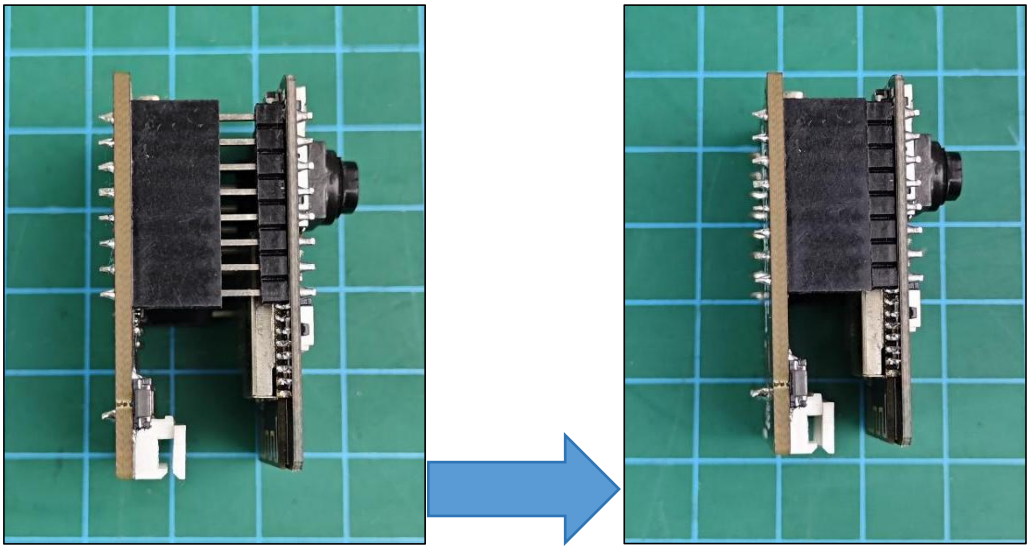


Interface	Instruction
TypeC serial port	Serial port communication, firmware programming
Pin header	Connect ESP32-Cam module.
Reset button	Press to restart the program.
I2C interface	Secondary development interface , connect with the master control.

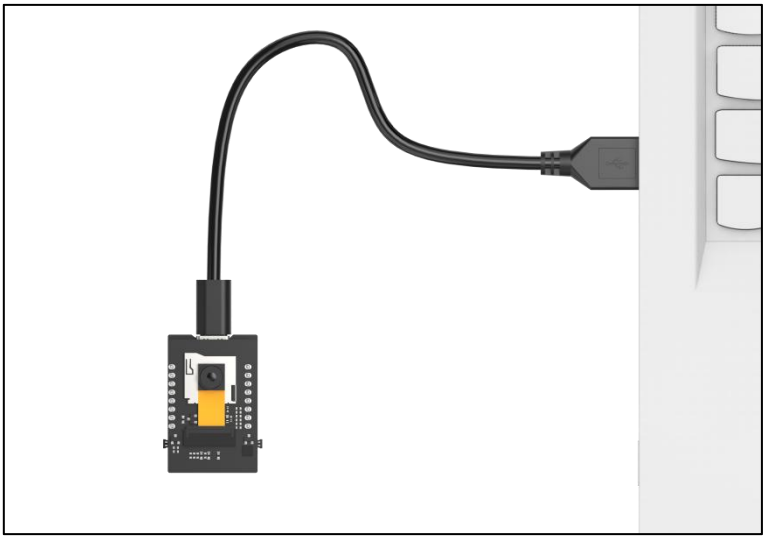
User button	User-defined extension development

2.2 Camera Installation and Connection

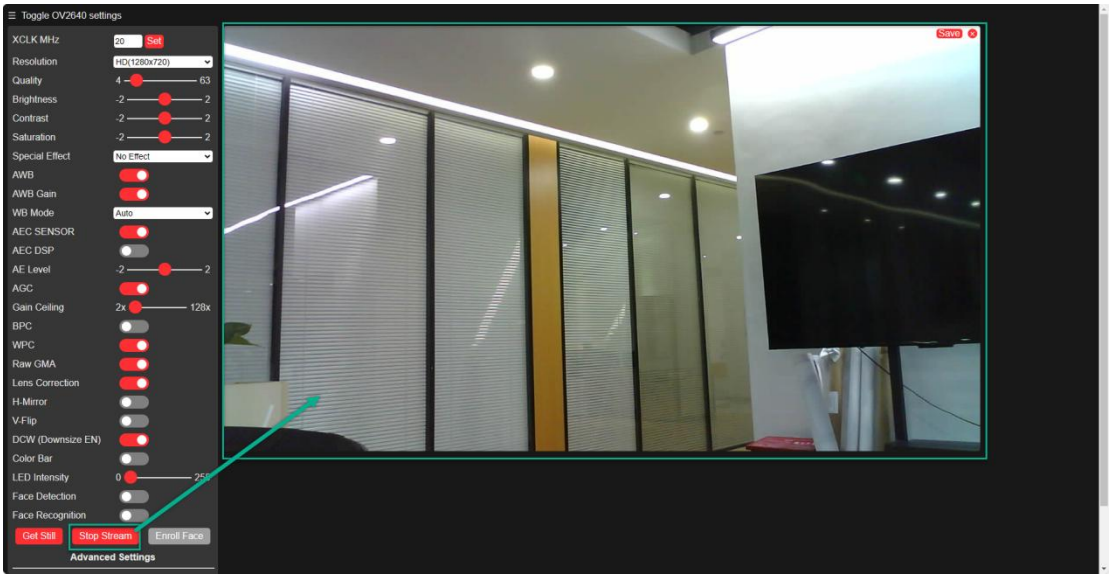
1) Align the pins of the ESP32-Cam with the pin headers on the camera expansion board.



2) Connect the camera to the computer with Type-C.



3) Refer to the “6.3 Live Camera Feed” tutorial to view the transmission interface.



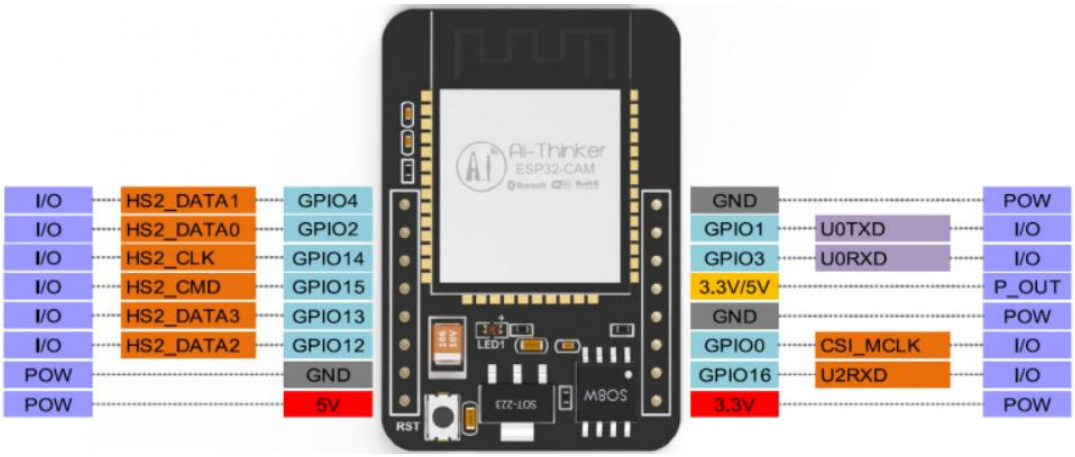
3. Note

1) Please ensure that the module is powered with a minimum of 5V 2A input to prevent potential image distortion.

2) The firmware of the live camera feed is already flashed before delivery. It is necessary to newly flash the program for other functions. Please refer to “1.Tutorials\6.AI Vision Game Course” to download the firmware.

4. Parameter Instruction

4.1 ESP32-Cam Pin Instruction



CAM	ESP32	SD	ESP32
D0	PIN5	CLK	PIN14
D1	PIN18	CMD	PIN15
D2	PIN19	DATA0	PIN2
D3	PIN21	DATA1/ LED	PIN4
D4	PIN36	DATA2	PIN12
D5	PIN39	DATA3	PIN13
D6	PIN34		
D7	PIN35		
XCLK	PIN0		
PCLK	PIN22		
VSYNC	PIN25		
HREF	PIN23		
SDA	PIN26		
SCL	PIN27		
POWER PIN	PIN32		

3.2 ESP32-Cam Parameter

Parameter	Explanation
Working Voltage	Powered by 5V 2A stabilized voltage supply.
CPU	Tensilica LX6 Dual Core Processor
Clock Rate	80 MHz (normal work speed), maximum is 240 MHz
SRAM	520 KB

External Storage	Support a maximum of 8 MB of PSRAM
Flash	Support up to 16 MB of external SPI Flash memory
WiFi Protocol	802.11 b/g/n/d/e/i/k/r, maximum speed is 150 Mbps
WiFi Frequency Range	2.4 ~ 2.5 GHz
Bluetooth Protocol	Support Bluetooth v4.2, including traditional Bluetooth (BR / EDR) and low-power Bluetooth (BLE)
Camera	Support OV2640 and OV7670 cameras with a built-in flash
Image Download	Support uploading images through Wi-Fi
Storage	Support microSD
Port	GPIO/ADC/ DAC/ SPI/ I2C/PWM/ UART
Power Supply Current	In deep-sleep method mode, minimum is 6mA
Working Method	STA/AP/STA+AP

Secondary Development	Support
Dimension	27 × 40.5 × 4.5 mm

Note: Product specifications are the theoretical values and are provided for reference only. Please rely on actual performance for accurate assessment.

You can access “4.Extension Learning\4.Sensors\ESP32-cam Camera” to check the detailed parameters in the Specification.

You can access “1.Tutorials\6.AI Vision Game Course” to check the program outcome of the ESP32-Cam camera.